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Effect of a ChatGPT-based digital counseling intervention on anxiety and depression in patients with cancer: A prospective, randomized trial

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ABSTRACT

Background: Psychological distress is prevalent among newly diagnosed cancer patients, often exacerbating treatment-related anxiety and depression. Artificial intelligence (AI)-driven interventions, such as large language models (LLMs), offer scalable solutions for patient education. However, their effectiveness in mitigating psychological distress in oncology remains unverified.

Methods: We conducted a prospective, two-center, randomized controlled trial between July and October 2024 to evaluate the effect of a ChatGPT-based digital counseling intervention on anxiety and depression in chemotherapy-naïve cancer patients. Participants were randomized 1:1 to either a ChatGPT-assisted counseling group or a standard clinician-led education group. The primary outcome was the change in Hospital Anxiety and Depression Scale (HADS) scores before the second chemotherapy cycle. Secondary outcomes included patient engagement behavior and the accuracy of AI-generated responses. Multivariable logistic regression assessed independent predictors of anxiety and depression reduction. The trial is registered with ClinicalTrials.gov (NCT06854315).

Findings: Of 196 screened patients, 160 were randomized (ChatGPT: n = 80; control: n = 80). The final cohort comprised 150 patients (median age: 64 years; 53.3 % female). At follow-up, the ChatGPT group exhibited significantly lower median HADS-Anxiety (6 [0–17] vs. 8 [1–17]; P = 0.002) and HADS-Depression (6 [0–20] vs. 9 [0–19]; P < 0.001) scores compared to controls. ChatGPT use was the strongest predictor of anxiety reduction (OR, 18.684; 95 % CI, 3.981–87.685; P < 0.001). AI responses were deemed appropriate in 88.5 % of cases, with inaccuracies most common in dietary and sexual health recommendations.

Interpretation: A ChatGPT-based counseling intervention significantly reduced anxiety and depression among newly diagnosed cancer patients prior to chemotherapy. These findings support the potential integration of AI-driven digital tools as adjuncts to conventional oncology education, enhancing patient support and engagement. Further research is needed to refine AI implementation in clinical practice.

1. Introduction

Newly diagnosed cancer patients frequently experience pronounced anxiety and uncertainty, driven by concerns about treatment-related side effects, disease prognosis, and overall quality of life [1,2]. In an effort to better understand their disease and mitigate uncertainty, a growing proportion of patients consult online platforms and social media to obtain information about their diagnosis and potential treatment strategies [2]. However, the credibility and accuracy of these platforms vary widely, and unregulated sources often present incomplete or misleading content [3,4]. Such misinformation not only compounds patients' emotional distress but can also undermine treatment adherence by fostering unrealistic expectations or unfounded fears [5].

Recent advancements in artificial intelligence (AI) offer promising alternatives for delivering more reliable and accessible patient education. In particular, large language models (LLMs) have garnered

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attention for their capacity to generate context-specific, user-friendly responses in real time [6,7]. Among these tools, the Chat Generative Pre-trained Transformer (ChatGPT) has become especially prominent, as it can engage patients in interactive "conversations" and tailor information to individual queries [8,9]. While AI-driven interventions such as Woebot have proven effective in mitigating psychological distress in certain populations [6], there remains a critical gap in understanding how ChatGPT might influence anxiety and depression among newly diagnosed cancer patients, who are especially vulnerable to the stress associated with upcoming chemotherapy.

Accordingly, this study investigates whether a ChatGPT-based digital counseling intervention can alleviate anxiety and depression in newly diagnosed cancer patients beginning anti-cancer treatment. By assessing the psychosocial impact of AI-mediated, on-demand information and support, we aim to elucidate the potential of LLMs to complement standard oncology education. Through this approach, we aim to enhance the psychosocial well-being of patients navigating the challenges of a cancer diagnosis.

2. Methods

2.1. Study design and randomization

This prospective, two-center, controlled study was conducted from July 2024 to October 2024 to evaluate the effects of a ChatGPT-based digital counseling intervention on anxiety and depression in patients newly diagnosed with solid tumors. Participants were randomized in a 1:1 ratio to either a ChatGPT arm or a standard arm using a computer-generated allocation system. Investigators responsible for outcome assessment were blinded to group assignments.

2.2. Eligibility criteria

Eligible participants were adults (≥18 years) with newly diagnosed, treatment-naive solid tumors, an Eastern Cooperative Oncology Group (ECOG) performance status < 3, and sufficient cognitive capacity to complete study questionnaires. Patients with recurrent disease, cranial metastases affecting cognitive function, or ECOG ≥ 3 were excluded. Written informed consent was obtained from all participants.

2.3. Interventions

Standard Arm: Participants received routine chemotherapy education provided by the treating oncology physician and chemotherapy nurse.

ChatGPT Arm: In addition to standard education, participants could submit up to 10 questions to the ChatGPT-4.0 model regarding their treatment or any cancer related concerns. Two oncologists (O.Y. and O. S.) reviewed each AI-generated response for accuracy, classifying it as (1) "appropriate," (2) "inappropriate," or (3) "unreliable." When responses were deemed inappropriate or unreliable, corrected information was subsequently provided.

2.4. Outcome measures

The primary outcomes were changes in anxiety and depression, assessed using the Hospital Anxiety and Depression Scale (HADS). This 14-item instrument includes subscales for anxiety (HADS-A) and depression (HADS-D), each ranging from 0 to 21 [10,11]

2.5. Data collection

At baseline (before any educational intervention), demographic and clinical information—including age, sex, educational level, cancer type, and disease stage—was recorded for each participant. Anxiety and depression were assessed using the Hospital Anxiety and Depression Scale (HADS), yielding HADS-A1 and HADS-D1 scores. Prior to the second chemotherapy cycle, HADS-A2 and HADS-D2 scores were obtained to evaluate changes over time.

All queries submitted to the ChatGPT-4.0 platform were automatically logged for subsequent content analysis. Additionally, the topics addressed during in-person education sessions were compared with the questions patients submitted to ChatGPT to assess differences in information-seeking behavior.

Details regarding participant flow and attrition are provided in the Results section, along with a CONSORT diagram illustrating the enrollment and follow-up process.

2.6. Ethical considerations

This study was approved by the Gazi University Faculty of Medicine Ethics Committee (E-1096068) and conducted in accordance with the principles of the Declaration of Helsinki.

2.7. Statistical analysis

A priori power analysis was conducted to determine the required sample size for detecting a clinically meaningful difference in HADS-Anxiety and HADS-Depression scores. Assuming a 2-point difference in HADS-A scores, with an alpha of 0.05 and a power of 80 %, a total of 68 participants per group were required. To account for potential attrition, we aimed to recruit 80 participants per arm, leading to a total sample size of 160.

All statistical analyses were performed using SPSS (IBM SPSS Statistics, version 27.0). Continuous variables were reported as median (minimum–maximum) or mean ± standard deviation, depending on the distribution of the data. Categorical variables were presented as frequency (n) and percentage (%).

The Kolmogorov-Smirnov test was used to assess the normality assumption for continuous variables. For data not demonstrating a normal distribution, the Mann-Whitney *U* test was used to compare groups, whereas the chi-square test was applied for categorical variables.

Changes over time within each group were analyzed using the nonparametric Wilcoxon signed-rank test for paired data. To determine the likelihood of a decrease in HADS scores, a multivariable logistic regression analysis was conducted, controlling for potential confounding factors such as age, sex, educational level, type of malignancy, and treatment intent. The effects of the variables included in the regression model were expressed as odds ratios (ORs) with 95 % confidence intervals (CIs).

A Minimal Clinically Important Difference (MCID) of 1.5 points or more in the HADS anxiety score was considered clinically meaningful, based on prior validation studies in similar clinical settings [11,12].

3. Results

3.1. Patient enrollment and baseline characteristics

Between July 2024 and October 2024, 196 patients were assessed for eligibility at two oncology centers in Türkiye; 36 declined consent, leaving 160 who were randomly assigned (1:1) to the ChatGPT or control arms. During the study, 10 participants (5 in each arm) were lost to follow-up or withdrew for various reasons, such as chemotherapy-related complications or transfer of care to another institution. Thus, 150 patients ultimately completed the study (Figure 1).

The median age of the final cohort was 64 years (range, 30–83), and 53.3 % were female. Baseline demographic and clinical characteristics, including cancer type, disease stage, and educational level, were comparable between the ChatGPT and control groups ($p > 0.05$). A detailed summary of these characteristics is provided in Table 1.

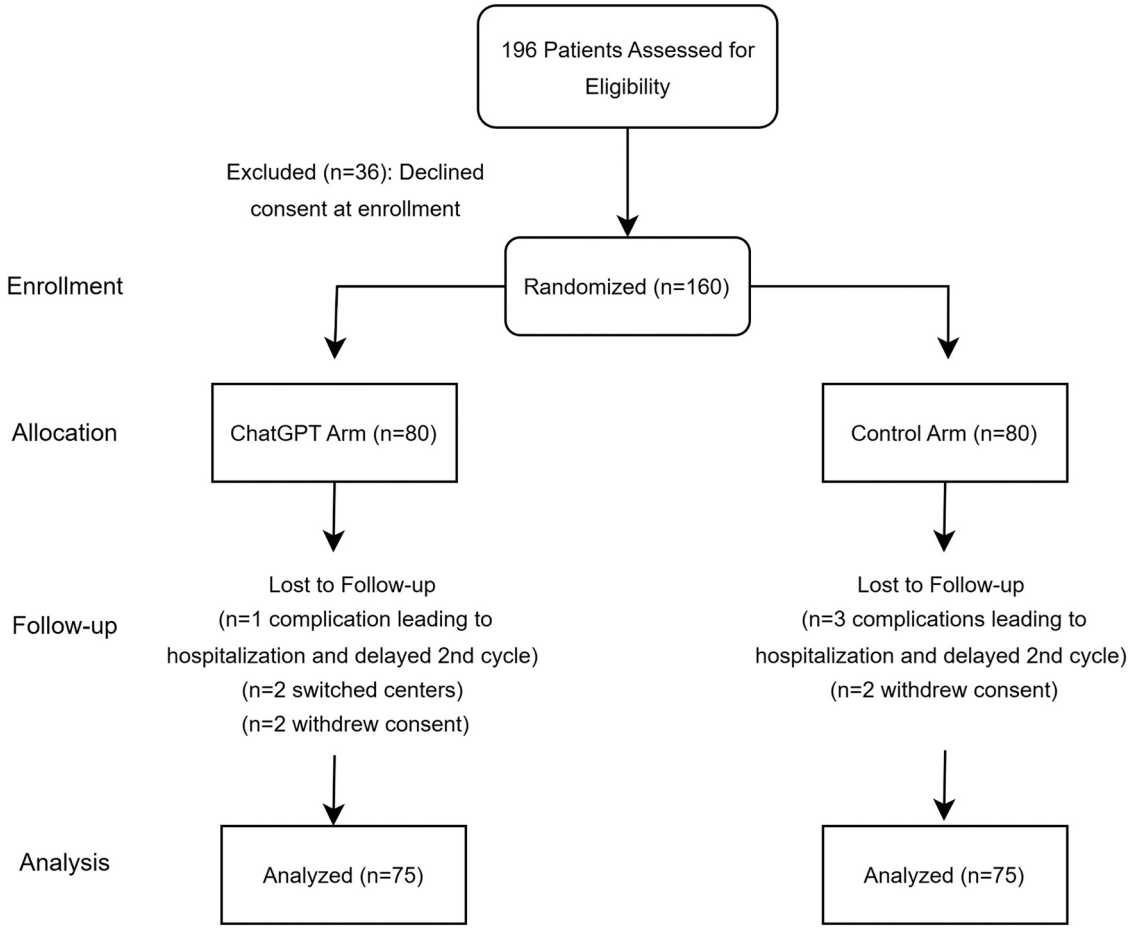

Fig. 1. Consort diagram.

3.2. Anxiety and depression outcomes

At baseline, median HADS-Anxiety (HADS-A) scores were 7 (range, 0–20) in the ChatGPT arm and 6 (range, 0–16) in the control arm ($p = 0.086$), while median HADS-Depression (HADS-D) scores were 6 (range, 0–20) vs 5 (range, 1–14), respectively ($p = 0.297$). By the time of the second chemotherapy cycle, the ChatGPT arm demonstrated significantly lower HADS-A and HADS-D scores than the control group—6 (range, 0–17) vs 8 (range, 1–17) ($p = 0.002$) for anxiety and 6 (range, 0–20) vs 9 (range, 0–19) ($p < 0.001$) for depression. Within-group analyses showed a significant decrease in HADS-A from baseline to the second cycle in the ChatGPT arm ($p = 0.009$), whereas HADS-D remained stable ($p = 0.163$). In contrast, the control arm exhibited significant increases in both HADS-A ($p < 0.001$) and HADS-D ($p < 0.001$).

In multivariable logistic regression, ChatGPT use was the sole significant predictor of decreased HADS-A (OR, 18.684; 95 % CI, 3.981–87.685; $p < 0.001$) (Table 2). For depression, ChatGPT use (OR, 4.274; $p = 0.010$) and postgraduate education (OR, 9.226; $p = 0.004$) were both independently associated with lower HADS-D scores, while other variables (including age, sex, disease stage, and malignancy type) did not show statistically significant effects(Table 3).

3.3. Patient queries and ChatGPT response accuracy

In the ChatGPT arm, the 20 most frequently asked chemotherapy-related questions are presented in Table 4, covering topics such as side effect management, nutrition, treatment protocols, and complementary therapies. The classification of ChatGPT responses—categorized as "appropriate," "inappropriate," or "unreliable"—is detailed in Table 5.

Among the 750 total patient inquiries, 664 (88.5 %) were classified as appropriate, while 73 (9.7 %) were categorized as unreliable, and 14 (1.8 %) were deemed inappropriate due to inaccuracies or lack of clinical relevance. The most common errors were observed in dietary recommendations (20 % of nutrition-related responses) and sexual health guidance (60 % of responses in this category). Specifically, unreliable dietary responses often provided overly generalized recommendations that did not account for chemotherapy-induced dietary restrictions. Similarly, responses regarding sexual activity lacked critical details on bleeding risk, fertility concerns, and medication excretion in bodily fluids, resulting in their classification as unreliable or inappropriate. These findings highlight the importance of expert oversight when integrating AI-based tools into patient education, particularly for topics requiring individualized clinical judgment.

Additionally, a supplementary survey indicated that 30 % of participants in the ChatGPT arm used the platform at home between chemotherapy cycles. In contrast, none of the participants in the control arm reported ChatGPT use outside of standard clinical care.

3.4. Comparison of In-Person vs ChatGPT queries

While patients did not ask their physician or nurse about sexual health during in-person education, 27 % of participants in the ChatGPT arm raised such concerns via the AI platform. Similarly, 30 % of patients did not inquire about the use of herbal supplements or vitamins during in-person education but sought information on these topics through ChatGPT.

Table 1
Baseline demographic and clinical characteristics of patients.

Characteristic	ChatGPT Arm (n = 75)	Control Arm (n = 75)	All Patients (N = 150)	p-value
Age, years median (min-max)	63 (30–80)	65 (32–83)	64 (30–83)	0813
Sex, n (%)				0.252
Female:	36 (48.0 %)	44 (58.7 %)	80 (53.3 %)	
Male:	39 (52.0 %)	31 (41.3 %)	70 (46.7 %)	
Cancer type, n (%)				0.270
- Breast	23 (30.7 %)	15 (20.0 %)	38 (25 %)	
- Ovarian	4 (5.3 %)	17 (22.7 %)	21 (14 %)	
- Colorectal	10 (13.3 %)	15 (20.0 %)	25 (17 %)	
- NSCLC	17 (22.7 %)	7 (9.3 %)	24 (16 %)	
- SCLC	4 (5.3 %)	0 (0 %)	4 (3 %)	
- Testicular	2 (2.7 %)	1 (1.3 %)	3 (2 %)	
- Endometrial	1 (1.3 %)	9 (12.0 %)	10 (7 %)	
- Laryngeal	5 (6.7 %)	0 (0 %)	5 (3 %)	
- Bladder	1 (1.3 %)	2 (2.7 %)	3 (2 %)	
- Gastric	3 (4.0 %)	5 (6.7 %)	8 (5 %)	
- Pancreatic	5 (6.7 %)	4 (5.3 %)	9 (6 %)	
Disease stage, n (%)				0.635
- Stage I	1 (1.3 %)	2 (2.7 %)	3 (2.0 %)	
- Stage II	21 (28.0 %)	16 (21.3 %)	37 (24.6 %)	
- Stage III	17 (22.7 %)	29 (38.7 %)	46 (30.6 %)	
- Stage IV	36 (48.0 %)	28 (37.3 %)	64 (42.6 %)	
Treatment intent, n (%)				0.187
- Curative (adjuvant/ neoadjuvant)	39 (52.0 %)	47 (62.7 %)	86 (57 %)	
- Palliative (metastatic)	36 (48.0 %)	28 (37.3 %)	64 (43 %)	
Educational Level, n (%)				0.550
Primary School or Lower	20 (26.7 %)	16 (21.3 %)	36 (24.0 %)	
High School	23 (30.6 %)	25 (33.3 %)	48 (32.0 %)	
University (Bachelor's)	20 (26.7 %)	26 (34.7 %)	46 (30.7 %)	
Postgraduate	12 (16.0 %)	8 (10.7 %)	20 (13.3 %)	

*Abbreviations: NSCLC, Non–Small Cell Lung Cancer; SCLC, Small Cell Lung Cancer.

Table 2
Logistic regression results for factors associated with a decrease in HADS-A scores.

Variable	OR (95 % CI)	p-value
Group		
Control (Ref)	—	—
ChatGPT	**18.684 (3.981–87.685)**	**< 0.001**
Treatment Type		
Palliative (Ref)	—	—
Curative	1.121 (0.309–4.060)	0.862
Stage		
Stage III/IV (Ref)	—	—
Stage I/II	0.579 (0.156–2.153)	0.415
Sex		
Male (Ref)	—	—
Female	0.499 (0.126–1.973)	0.321
Malignancy Type		
Other (Ref)	—	—
Breast	0.574 (0.106–3.100)	0.519
NSCLC	0.385 (0.075–1.982)	0.254
Gynecologic	0.176 (0.021–1.452)	0.107
Colorectal	0.411 (0.060–2.839)	0.367
Educational Level		
Primary or Lower (Ref)	—	—
High School	0.869 (0.219–3.441)	0.841
University (Bachelor's)	1.853 (0.454–7.569)	0.390
Postgraduate	1.847 (0.385–8.871)	0.443

*Abbreviations: NSCLC, Non–Small Cell Lung Cancer

Table 3
Logistic regression results for factors associated with a decrease in HADS-D scores.

Variable	OR (95 % CI)	p-value
Group		
Control (Ref)	—	—
ChatGPT	**4.274 (1.410–12.958)**	**0.010**
Treatment Type		
Palliative (Ref)	—	—
Curative	1.121 (0.309–4.060)	0.862
Stage		
Stage III/IV (Ref)	—	—
Stage I/II	0.579 (0.156–2.153)	0.415
Sex		
Male (Ref)	—	—
Female	0.499 (0.126–1.973)	0.321
Malignancy Type		
Other (Ref)	—	—
Breast	0.574 (0.106–3.100)	0.519
NSCLC	0.385 (0.075–1.982)	0.254
Gynecologic	0.176 (0.021–1.452)	0.107
Colorectal	0.411 (0.060–2.839)	0.367
Educational Level		
Primary or Lower (Ref)	—	—
High School	1.315 (0.310–5.577)	0.710
University (Bachelor's)	3.421 (0.827–14.150)	0.090
Postgraduate	**9.226 (2.012–42.302)**	**0.004**

*Abbreviations: NSCLC, Non–Small Cell Lung Cancer

Table 4
Most frequently asked questions to ChatGPT, by topic.

Rank	Topic	Frequency (%)
1	Side Effects and Their Management	90
2	Nutrition and Diet	87
3	Chemotherapy Protocols and Treatment Process	83
4	Alternative Therapies and Herbal Supplements	72
5	Hygiene, Infection, and Sterilization	70
6	Psychological Support and Mental Health	68
7	Exercise, Physical Activity, and Fatigue	64
8	Social Life, Daily Activities, and Responsibilities	59
9	Drug Interactions, Comorbidities, and Combined Medications	51
10	Sexual Life and Future Fertility	27

*Note: Percentages represent the proportion of patients who posed questions on each topic among the ChatGPT arm (n = 75).

4. Discussion

This study represents the first known investigation into the effects of a large language model (LLM)–based artificial intelligence tool on the anxiety and depression level of cancer patients. Our findings revealed that patients in the ChatGPT arm exhibited significantly lower anxiety and depression scores prior to the second chemotherapy cycle than those in the control arm.These results suggest that ChatGPT has the potential to alleviate patient concerns, address informational gaps, and provide psychological support during a highly stressful phase of treatment.

Our findings underscore the importance of effective patient education in mitigating treatment-related distress. In our study, the significant increases in both anxiety and depression scores observed in the control arm may be attributed, in part, to an information deficit that amplified patients' perceptions of treatment-related side effects and heightened their fears regarding chemotherapy. In the absence of supplemental counseling, patients in the control arm might have experienced greater uncertainty and interpreted common adverse effects as more severe, thereby exacerbating their emotional distress. These factors likely contributed to the observed deterioration in psychological outcomes in this group compared to the experimental arm, which benefited from timely, personalized digital counseling [4,13].

Table 5
Frequently asked questions about chemotherapy.

Questions	Responds	Responds Evaluation	Rate of Questions Asked to ChatGPT by Patients n (%)
What are the most common side effects of chemotherapy, and how can they be managed?	Common side effects of chemotherapy include nausea, vomiting, fatigue, hair loss, infections, and changes in appetite. Management strategies involve supportive medications, dietary adjustments, and adequate hydration.	appropriate	75 %
What dietary considerations should be taken into account during chemotherapy, and are there any food restrictions?	A balanced diet rich in proteins, vitamins, and minerals is recommended. Patients should avoid raw or undercooked foods due to infection risks and should limit alcohol and caffeine consumption.	unreliable	70 %
How does chemotherapy affect the immune system, and what precautions can be taken to reduce the risk of infection?	Chemotherapy can weaken the immune system, increasing infection risk. Preventive measures include good hygiene, avoiding crowded places, and receiving recommended vaccinations.	appropriate	65 %
How long do nausea and vomiting caused by chemotherapy last, and what measures can be taken to control these symptoms?	Nausea and vomiting can last for several days after chemotherapy but can be controlled with antiemetic medications, hydration, and small, frequent meals.	appropriate	60 %
When does chemotherapy-induced hair loss begin, and what is the process of hair regrowth after treatment?	Hair loss typically begins 2–3 weeks after starting chemotherapy and regrowth occurs within 3–6 months after treatment completion.	appropriate	55 %
Can daily activities be continued during chemotherapy, and are there any restrictions on exposure to public spaces?	Patients can generally continue daily activities but should avoid crowded places and ensure proper hygiene to reduce infection risks.	appropriate	55 %
Is the use of vitamin and dietary supplements safe during chemotherapy?	Some supplements may interfere with chemotherapy. Patients should consult their physician before using any vitamins or herbal products.	appropriate	40 %
Is it appropriate to use gastroprotective agents, pain relievers, or other prescribed medications during chemotherapy?	Certain medications can be used safely, but it is important to consult an oncologist before starting any new drug.	appropriate	38 %
What are the long-term side effects of chemotherapy, and which organs may be affected?	Long-term effects may include neuropathy, cognitive impairment, fertility issues, and cardiovascular risks. Monitoring and supportive care are important.	appropriate	36 %
Is cancer a genetic disease, and what is the risk of transmission to family members?	While some cancers have genetic components, most are influenced by lifestyle and environmental factors. Genetic counseling may be useful for hereditary cancer risk assessment.	appropriate	35 %
Is it safe to drive after receiving chemotherapy?	Patients may experience dizziness and fatigue post-chemotherapy, so driving should be avoided on treatment days and assessed based on individual tolerance.	appropriate	30 %
Is it safe to continue anticoagulant therapy or other regularly prescribed medications during chemotherapy	Many medications can be safely continued, but interactions with chemotherapy should be reviewed by an oncologist.	appropriate	23 %
Is it safe to dye hair, use cosmetic products, or continue personal care practices after chemotherapy?	Hair dyeing should be postponed until hair regrowth stabilizes. Mild and hypoallergenic cosmetic products are recommended.	appropriate	22 %
How is the necessity of additional treatments determined after chemotherapy?	Additional treatments—including radiotherapy, immunotherapy, targeted, or hormonal therapies—are determined by cancer type, stage, and response to chemotherapy. Oncologists use clinical assessments, imaging, and biomarkers to guide further treatment decisions.	appropriate	21 %
Is it safe for chemotherapy patients to keep pets at home?	Yes, but precautions such as proper hand hygiene and avoiding direct contact with pet waste should be taken to reduce infection risk.	appropriate	18 %
Is it safe to engage in sexual activity during chemotherapy, and does it pose any risk to the partner?	Sexual activity is generally safe, but protection is advised due to possible drug excretion in bodily fluids.	unreliable	18 %
What is the risk of cancer recurrence after chemotherapy?	The recurrence risk depends on cancer type, stage, and response to treatment. Regular follow-ups and screenings are essential for early detection.	appropriate	16 %
Which symptoms before chemotherapy may contraindicate treatment initiation?	Severe infections, uncontrolled fever, extreme weakness, or critical organ dysfunction may require treatment delay or modifications.	appropriate	12 %
Is weight loss or weight gain	Weight changes vary among patients. Proper	appropriate	10 %

(*continued on next page*)

Table 5 (*continued*)

Questions	Responds	Responds Evaluation	Rate of Questions Asked to ChatGPT by Patients n (%)
expected during chemotherapy, and what measures can be taken to prevent it?	nutrition and dietary guidance can help maintain a stable weight.		
What is the likelihood of complete remission after chemotherapy?	The chance of remission depends on cancer type, stage, and treatment response. Oncologists provide individualized prognostic assessments.	appropriate	9 %

Our study further identified a significant association between lower educational attainment and an increased risk of depression among cancer patients, a finding that aligns with existing literature. This finding aligns with existing literature. For instance, a systematic review highlighted that patients with lower educational levels exhibit higher rates of depressive and anxiety symptoms [14]. Similarly, a study investigating factors associated with depressive symptoms among cancer patients found that lower educational status was significantly linked to higher depression scores [15]. These trends may be attributed to limited health literacy among individuals with lower education, leading to challenges in understanding disease processes and treatment options, thereby increasing psychological distress. Conversely, higher educational attainment may enhance coping mechanisms, as these individuals are better equipped to comprehend medical information and navigate healthcare systems [14,15]. Therefore, implementing targeted educational and psychosocial interventions for cancer patients with lower educational backgrounds could be crucial in mitigating depression risk and improving overall mental health outcomes.

The use of AI in healthcare has expanded rapidly in recent years, with applications ranging from decision support systems to psychological interventions [7]. AI-driven platforms such as Woebot and Bravemind have demonstrated efficacy in managing stress, anxiety, and post-traumatic stress disorder (PTSD) through techniques like Cognitive Behavioral Therapy (CBT) and Virtual Reality Exposure Therapy (VRET) [6,16]. Consistent with these advancements, our study highlights the potential utility of AI-based tools in the pretreatment phase, suggesting that they can play a pivotal role in reducing anxiety and depression even before therapy begins.

In face-to-face clinical encounters, patients often feel nervous or intimidated when speaking with their physicians, which can lead many to withhold questions. As a result, many refrain from asking all of their questions. In contrast, online systems like ChatGPT offer a less intimidating and more accessible environment, enabling patients to seek information without the pressure of direct interaction. Furthermore, ChatGPT provides a degree of psychosocial support by delivering empathetic responses that help patients feel understood and at ease [17]. This dynamic not only facilitates the disclosure of concerns that might otherwise remain unexpressed but also complements conventional medical consultations by enhancing patient engagement and treatment adherence [3,18]. The findings of our study extend this evidence base, suggesting that ChatGPT's empathetic and interactive design can complement traditional psychosocial support.

In addition to general online searches, social media platforms serve as important sources of information and social support for cancer patients. However, the content on these platforms is often unregulated and may be driven by commercial interests or the pursuit of popularity, leading to the dissemination of unverified or misleading information [4,13]. Newly diagnosed cancer patients often turn to the internet and social media in search of guidance, but these platforms frequently prioritize user metrics rather than factual accuracy. Content creators may present information selectively or exaggerate claims to attract attention, making it difficult for patients to discern reliable sources. Additionally, information is typically presented from the perspective of the content provider rather than being tailored to the individual needs of the patient. In contrast, AI-based tools like ChatGPT allow patients to directly ask the questions that concern them and receive responses that align with their specific queries [3]. Furthermore, ChatGPT adapts its responses based on the user's phrasing and level of understanding, ensuring accessibility for individuals with varying educational backgrounds. Given its generally high accuracy in medical responses, ChatGPT provides a structured and personalized approach to information-seeking, enabling patients to obtain rapid and reliable answers to their concerns [8,19].

Despite its potential, the use of ChatGPT in healthcare settings is not without challenges. The model lacks access to individual clinical histories and cannot perform physical examinations or diagnostic tests, which limits its ability to provide comprehensive guidance [20]. Additionally, because ChatGPT relies on a training dataset that may not include the most recent medical guidelines and advancements, there is a risk of providing outdated or inaccurate information [21,22]. In our study, we attempted to mitigate this concern by instituting specialist oversight for all AI-generated responses; however, real-time clinical review may not be feasible in many practical scenarios, warranting caution in fully automating such interactions. Moreover, the growing reliance on such tools could inadvertently weaken the patient–physician relationship by reducing direct interactions [9,23]. Therefore, ChatGPT should be integrated as a complementary tool, emphasizing the need for expert oversight to ensure its appropriate use in clinical practice. A potential strategy to enhance its reliability could be the development of additional interfaces that enable ChatGPT to incorporate international, evidence-based patient guidelines from reputable organizations such as ESMO and ASCO. Such an approach may improve the accuracy and relevance of the information provided, ensuring that patients receive guidance aligned with the latest medical standards.

A key strength of this study is its innovative focus on the use of LLMs like ChatGPT during the pretreatment phase, thereby addressing an underexplored aspect of AI in healthcare. The randomized design and inclusion of a diverse patient population enhance the generalizability of our findings. However, the study has limitations, including the reliance on self-reported outcomes, which may be influenced by response bias. Additionally, the short follow-up period limits the assessment of long-term psychological benefits. Future studies should explore the sustained impact of ChatGPT and its integration into broader psychosocial support strategies.

5. Conclusion

This study demonstrates that a ChatGPT-based digital counseling application can significantly reduce anxiety and depression scores among patients with cancer during the pre-chemotherapy period. These findings suggest that digital interventions, such as ChatGPT, may serve as valuable adjuncts or alternatives to conventional psychosocial support methods. The secure and widespread integration of AI tools like ChatGPT into clinical practice holds promise for alleviating the psychosocial burden experienced by oncology patients.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data Availability

Deidentified individual participant data (including data dictionaries) will be made available upon reasonable request to the corresponding author. The data will be shared for 12 months following publication, to researchers who provide a methodologically sound proposal.

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